

LWESHAKJ MICROSOFT EXCEL DETAILED TEACHING NOTES

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1. INTRODUCTION TO MICROSOFT EXCEL

What is Microsoft Excel?

Microsoft Excel is a spreadsheet application used to:

- Store data
- Organise information
- Perform calculations
- Analyse data
- Create reports
- Create charts
- Automate calculations using formulas and functions

Excel is commonly used in:

- Schools
- Businesses
- Banks
- Hospitals
- Government offices
- Accounting
- Inventory management
- Research
- Personal budgeting

Example

A school can use Excel to store:

Student	Gender	Class	Mathematics	English	Science
John	Male	Form 2A	75	68	82
Mary	Female	Form 2A	84	77	79
Peter	Male	Form 2B	56	61	63

Excel can then calculate each student's:

- Total
- Average
- Grade

- Position
 - Pass/Fail status
-

2. WORKBOOK AND WORKSHEET

Workbook

A workbook is the Excel file itself.

For example:

`Student_Results.xlsx`

A workbook can contain several worksheets.

Worksheet

A worksheet is an individual spreadsheet inside a workbook.

For example:

Workbook: `School_Results.xlsx`

It could contain:

- Form 1
- Form 2
- Form 3
- Form 4

Important distinction

Workbook = entire Excel file

Worksheet = one sheet inside the file

3. ROWS, COLUMNS AND CELLS

Rows

Rows run horizontally.

They are identified by numbers:

1, 2, 3, 4, 5...

Columns

Columns run vertically.

They are identified by letters:

A, B, C, D...

After Z:

AA, AB, AC...

Cell

A cell is the intersection of a row and a column.

For example:

B5

means:

- Column B
- Row 5

Cell Address

A cell address identifies the location of a cell.

Examples:

A1

B5

D20

AA100

4. ENTERING DATA

Excel can contain different types of data.

Text

John
Tanzania
Form 2A
Male

Numbers

25
100
75.5
2026

Dates

25/08/2026

Formulas

=A1+B1

Functions

=SUM(A1:A10)

5. CELL REFERENCES

A cell reference tells Excel where to obtain information.

Suppose:

A	B
Student	Score

A B
John 75

John's score is located in:

B2

Therefore:

=B2

returns:

75

6. EXCEL OPERATORS

Operators tell Excel what operation to perform.

Arithmetic operators

Operator	Meaning	Example
+	Addition	=A1+B1
-	Subtraction	=A1-B1
*	Multiplication	=A1*B1
/	Division	=A1/B1
^	Power	=A1^2
%	Percentage	=A1*10%

Example

If:

A1 = 20
B1 = 5

Then:

=A1+B1

returns:

25

while:

=A1*B1

returns:

100

7. COMPARISON OPERATORS

Comparison operators are extremely important when working with IF, COUNTIF, SUMIF, etc.

Operator	Meaning
=	Equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to
<>	Not equal to

Example

=A1>=50

If A1 contains 70, the result is:

TRUE

If A1 contains 40:

FALSE

8. BASIC FORMULAS

A formula normally begins with:

=

Addition

=A1+B1

Subtraction

=A1-B1

Multiplication

=A1*B1

Division

=A1/B1

Percentage

Suppose:

Raw score = 75

Maximum score = 100

Formula:

=75/100*100

Result:

75%

More practically:

=B2/C2*100

9. SUM FUNCTION

SUM adds numbers together.

Syntax

=SUM(number1,number2,...)

or:

=SUM(range)

Example

=SUM (B2 : B6)

If B2:B6 contains:

10
20
30
40
50

Result:

150

Why use SUM instead of:

=B2+B3+B4+B5+B6

Because:

=SUM (B2 : B6)

is easier to maintain when the dataset becomes larger.

10. AVERAGE

Calculates the arithmetic mean.

=AVERAGE (B2 : B10)

If the scores are:

60
70
80
90
100

Average:

80

11. MIN AND MAX

MIN

Finds the smallest value.

`=MIN (B2 : B100)`

MAX

Finds the largest value.

`=MAX (B2 : B100)`

Example:

45
78
91
63
52
`=MIN (A1 : A5)`

returns:

45

while:

`=MAX (A1 : A5)`

returns:

91

12. COUNT, COUNTA AND COUNTBLANK

COUNT

Counts cells containing numbers.

`=COUNT (A1 : A100)`

COUNTA

Counts non-empty cells.

=COUNTA (A1 : A100)

COUNTBLANK

Counts empty cells.

=COUNTBLANK (A1 : A100)

Important difference

Suppose:

A

John

75

Mary

82

Peter

=COUNT (A1 : A5)

returns:

2

because only 75 and 82 are numbers.

But:

=COUNTA (A1 : A5)

returns:

5

because all five cells contain something.

13. RELATIVE REFERENCES

A relative reference changes when a formula is copied.

Suppose:

=B2+C2

is entered in D2.

When copied to D3, Excel automatically changes it to:

=B3+C3

This is a relative reference.

14. ABSOLUTE REFERENCES

An absolute reference does not change when copied.

Example:

=B\$1

The \$ symbol locks the column and row.

Example

Suppose:

A	B
Product	Price
Pen	100
Book	500
Bag	2000

And tax rate is stored in:

D1 = 18%

Formula:

=B2*\$D\$1

When copied down, \$D\$1 remains fixed.

15. MIXED REFERENCES

There are three important reference types.

Relative

A1

Absolute

\$A\$1

Mixed

\$A1

or:

A\$1

Meaning

\$A1

- Column A is locked
- Row can change

A\$1

- Column can change
- Row 1 is locked

16. IF FUNCTION

IF makes a decision.

Syntax

=IF(logical_test,value_if_true,value_if_false)

Example

=IF (B2>=50, "PASS", "FAIL")

If B2 is 65:

PASS

If B2 is 40:

FAIL

Real-world scenario

A teacher wants Excel to determine whether students passed.

=IF (C2>=50, "PASS", "FAIL")

17. NESTED IF

A nested IF places one IF inside another.

Suppose grades are:

Score	Grade
80–100	A
70–79	B
60–69	C
50–59	D
Below 50	F

Formula:

=IF (B2>=80, "A", IF (B2>=70, "B", IF (B2>=60, "C", IF (B2>=50, "D", "F"))))

Excel evaluates the conditions from left to right.

18. IFS

IFS can make multiple-condition formulas easier to read.

=IFS (B2>=80, "A", B2>=70, "B", B2>=60, "C", B2>=50, "D", B2<50, "F")

Nested IF vs IFS

Nested IF:

```
=IF (B2>=80, "A", IF (B2>=70, "B", IF (B2>=60, "C", "F"))) )
```

IFS:

```
=IFS (B2>=80, "A", B2>=70, "B", B2>=60, "C", B2<60, "F")
```

For many conditions, IFS is generally easier to read.

19. AND

AND checks whether all conditions are TRUE.

```
=AND (A1>=50, B1>=50)
```

Both must be true.

Combined with IF

```
=IF (AND (B2>=50, C2>=50), "PASS", "FAIL")
```

This means:

The student passes only if both subjects are at least 50.

20. OR

OR checks whether at least one condition is TRUE.

```
=OR (A1>=50, B1>=50)
```

Combined with IF

```
=IF (OR (B2>=80, C2>=80), "EXCELLENT", "NORMAL")
```

If either subject is 80 or above, Excel returns EXCELLENT.

21. NOT

NOT reverses TRUE/FALSE.

`=NOT (A1=10)`

If A1 is 10:

FALSE

If A1 is 20:

TRUE

22. COUNTIF

COUNTIF counts cells based on one condition.

Syntax

`=COUNTIF (range, criteria)`

Example

Count males:

`=COUNTIF (B2:B100, "MALE")`

Count students who scored 50 or above:

`=COUNTIF (C2:C100, ">=50")`

Count students below 50:

`=COUNTIF (C2:C100, "<50")`

23. COUNTIFS

COUNTIFS handles multiple conditions.

Syntax

`=COUNTIFS (range1,criterial,range2,criteria2,...)`

Example

Count males in Form 2A:

`=COUNTIFS (B2:B100,"MALE",C2:C100,"Form 2A")`

This means:

Count rows where Gender = MALE AND Class = Form 2A.

Important concept

`COUNTIF` → one condition

`COUNTIFS` → multiple conditions

24. SUMIF

`SUMIF` adds values based on one condition.

Syntax

`=SUMIF (range,criteria,sum_range)`

Example:

Student	Gender	Score
John	Male	70
Mary	Female	80
Peter	Male	60

Total male scores:

`=SUMIF (B2:B4,"Male",C2:C4)`

Result:

130

25. SUMIFS

SUMIFS adds values based on multiple conditions.

Example:

```
=SUMIFS (C2:C100, B2:B100, "MALE", D2:D100, "Form 2A")
```

This means:

Add scores where Gender is MALE and Class is Form 2A.

Critical difference

Notice the structure:

SUMIF:

```
=SUMIF (criteria_range, criteria, sum_range)
```

SUMIFS:

```
=SUMIFS (sum_range, criteria_range1, criterial1, criteria_range2, criteria2)
```

This difference is extremely important for you to learn.

26. AVERAGEIF

Calculates an average based on one condition.

```
=AVERAGEIF (B2:B100, "Male", C2:C100)
```

Meaning:

Calculate the average score of male students.

27. AVERAGEIFS

Calculates an average based on multiple conditions.

```
=AVERAGEIFS (C2:C100, B2:B100, "Male", D2:D100, "Form 2A")
```

Meaning:

Calculate the average score of male students in Form 2A.

28. XLOOKUP

One of the most useful modern lookup functions.

Basic syntax

```
=XLOOKUP (lookup_value, lookup_array, return_array)
```

Example:

Student ID	Name	Score
S001	John	78
S002	Mary	85
S003	Peter	62

If A2 contains:

S002

Use:

```
=XLOOKUP (A2, A5:A7, B5:B7)
```

Result:

Mary

29. XLOOKUP WITH NOT FOUND MESSAGE

You can specify what Excel should display if the value doesn't exist.

```
=XLOOKUP (A2, A5:A100, B5:B100, "NAME IS NOT FOUND")
```

If A2 doesn't exist, Excel returns:

NAME IS NOT FOUND

This is better than leaving the user with an error such as #N/A.

30. XLOOKUP — IMPORTANT CONCEPT

You often think XLOOKUP searches an entire table automatically.

It doesn't.

You explicitly tell Excel:

What to search for

A2

Where to search

A5:A100

What to return

B5:B100

So:

`=XLOOKUP(A2,A5:A100,B5:B100)`

means:

Find the value in A2 inside A5:A100 and return the corresponding value from B5:B100.

31. AMPERSAND &

The ampersand joins text together.

`=A2&" "&B2`

If:

A2 = Lwsha
B2 = Kelvin

Result:

Lwsha Kelvin

Combining text and numbers

= "Score: "&B2

If B2 is 85:

Score: 85

32. TEXT FUNCTIONS

LEFT

Returns characters from the left.

=LEFT (A2, 3)

If:

A2 = Tanzania

Result:

Tan

RIGHT

=RIGHT (A2, 3)

Result:

nia

MID

=MID (A2, 3, 4)

Means:

Start at character 3 and return 4 characters.

33. LEN

Counts characters.

`=LEN (A2)`

If A2 contains:

Excel

Result:

5

Spaces are also counted.

34. TRIM

Removes unnecessary spaces.

`=TRIM (A2)`

This is particularly useful when data has been copied from another system.

35. SUBSTITUTE

Replaces specific text.

`=SUBSTITUTE (A2, "Male", "M")`

For example:

MALE

can become:

M

36. TEXT FUNCTION

TEXT allows numbers and dates to be displayed in a specified format.

Example:

```
=TEXT (A2, "0.00")
```

If A2 is:

75

Result:

75.00

Date example:

```
=TEXT (A2, "dd/mm/yyyy")
```

37. ROUND

Rounds a number.

```
=ROUND (A2, 2)
```

If:

A2 = 75.6789

Result:

75.68

Other useful functions:

```
=ROUNDUP (A2, 2)
```

```
=ROUNDDOWN (A2, 2)
```

38. DATE FUNCTIONS

TODAY

Returns today's date.

=TODAY ()

NOW

Returns current date and time.

=NOW ()

YEAR

=YEAR (A2)

MONTH

=MONTH (A2)

DAY

=DAY (A2)

39. SORT

Sorts data using a formula.

=SORT (A2:C100)

Can also sort based on a particular column.

40. FILTER

One of the most powerful modern Excel functions.

Suppose you want only male students:

=FILTER (A2:D100, B2:B100="Male")

Excel returns all rows where Gender is Male.

Why FILTER is powerful

Traditional filtering changes the displayed dataset.

`FILTER()` creates a new dynamic result from your formula.

41. UNIQUE

Returns unique values.

`=UNIQUE(B2:B100)`

If the original data contains:

Form 2A
Form 2A
Form 2B
Form 2A
Form 2C
Form 2B

Result:

Form 2A
Form 2B
Form 2C

42. SEQUENCE

Generates sequential numbers.

`=SEQUENCE(10)`

Produces:

1
2
3
...
10

You can use it to generate:

- Student numbers
 - Serial numbers
 - Test data
 - Dates
 - Number sequences
-

43. SORT + FILTER + UNIQUE

These functions become extremely powerful when combined.

For example:

```
=SORT (UNIQUE (B2:B100) )
```

This produces a sorted list of unique values.

44. CONDITIONAL FORMATTING

Conditional formatting automatically changes the appearance of cells based on rules.

Examples:

- Scores below 50 → highlight
- Scores above 80 → highlight
- Duplicate names → highlight
- Highest values → highlight

Example rule:

```
Cell Value < 50
```

This can make failing students immediately visible.

45. DATA VALIDATION

Data Validation controls what users are allowed to enter.

For example, instead of allowing you/anyone to type:

Male
male
MALE
M
Males

you can create a dropdown:

Male
Female

This improves data consistency.

46. SORTING AND FILTERING

Sorting

Arrange data:

- $A \rightarrow Z$
- $Z \rightarrow A$
- Smallest $\rightarrow$ Largest
- Largest $\rightarrow$ Smallest

Filtering

Display only records meeting a condition.

Example:

Show only:

Form 2A

or:

Female

or:

Score ≥ 50

47. EXCEL TABLES

A normal range:

A1:F100

can be converted into an Excel Table.

Shortcut:

Ctrl + T

Tables provide:

- Automatic formatting
 - Filters
 - Automatic expansion
 - Structured references
 - Easier formulas
 - Easier data management
-

48. COMMON EXCEL ERRORS

You should understand errors rather than simply fear them.

#DIV/0!

Trying to divide by zero.

=A1/B1

where B1 = 0.

#N/A

A value cannot be found.

Common with:

XLOOKUP

#VALUE!

Wrong type of data used in a calculation.

#REF!

A formula refers to an invalid cell.

#NAME?

Excel doesn't recognise something in the formula.

#SPILL!

A dynamic-array formula cannot expand because something is blocking its output area.

49. IFERROR

IFERROR allows you to control what happens when a formula produces an error.

Example:

```
=IFERROR(A2/B2,0)
```

If the calculation works, Excel returns the result.

If an error occurs, Excel returns:

0

With XLOOKUP:

```
=IFERROR(XLOOKUP(A2,D2:D100,E2:E100),"NOT FOUND")
```

50. COMBINING FUNCTIONS

This is where you move from basic Excel to real Excel problem solving.

For example:

```
=IF(AVERAGE(B2:F2)>=50,"PASS","FAIL")
```

Excel first calculates:

AVERAGE (B2 : F2)

Then IF evaluates the result.

Another example:

=IF (AND (B2>=50 , C2>=50) , "PASS" , "FAIL")

This combines:

IF + AND

51. REAL-WORLD SCENARIO

Imagine a school has 100 students.

Columns:

Column	Information
A	Student ID
B	Student Name
C	Gender
D	Class
E	Mathematics
F	English
G	Science
H	Total
I	Average
J	Grade
K	Status

Total

=SUM (E2 : G2)

Average

=AVERAGE (E2 : G2)

Grade

=IFS (
I2>=80 , "A" ,
I2>=70 , "B" ,

```
I2>=60,"C",  
I2>=50,"D",  
I2<50,"F"  
)
```

Status

```
=IF(I2>=50,"PASS","FAIL")
```

Now the spreadsheet becomes a complete student-results system.

52. ADVANCED CHALLENGE

Test yourself:

How many male students in Form 2A scored at least 50 in Mathematics?

You need:

```
=COUNTIFS(C2:C101,"MALE",D2:D101,"Form 2A",E2:E101,">=50")
```

This is much better than simply wanting to memorise COUNTIFS.

You learn to translate a real-world question into Excel logic.

53. THE MOST IMPORTANT SKILL

The objective of Excel training should not be:

"I know 30 Excel functions."

It should be:

"I can look at a problem, identify the conditions and data involved, and construct the appropriate Excel solution."

For example:

Question:

How much did male students in Form 2A score altogether?

Student identifies:

What are we adding?

Scores.

What condition?

Gender = Male.

Second condition?

Class = Form 2A.

Therefore:

```
=SUMIFS (ScoreRange, GenderRange, "MALE", ClassRange, "Form 2A")
```

That's actual Excel understanding.

54. PROGRESSION FOR YOURSELF

You should learn the course in this order:

LEVEL 1 — FOUNDATION

- Excel interface
- Cells
- Rows/columns
- Data entry
- Formatting
- Basic arithmetic
- Basic functions

LEVEL 2 — FORMULAS

- Cell references
- Relative references
- Absolute references
- Mixed references
- Operators
- Percentages

LEVEL 3 — LOGIC

- IF
- Nested IF
- IFS
- AND
- OR
- NOT
- IFERROR

LEVEL 4 — DATA ANALYSIS

- COUNTIF
- COUNTIFS
- SUMIF
- SUMIFS
- AVERAGEIF
- AVERAGEIFS

LEVEL 5 — DATA RETRIEVAL

- XLOOKUP
- VLOOKUP
- INDEX
- MATCH
- INDEX + MATCH

LEVEL 6 — TEXT & DATA CLEANING

- LEFT
- RIGHT
- MID
- LEN
- TRIM
- CLEAN
- SUBSTITUTE
- TEXT
- &

LEVEL 7 — MODERN EXCEL

- FILTER
- SORT
- UNIQUE

- SEQUENCE
- Dynamic arrays

LEVEL 8 — DATA MANAGEMENT

- Tables
- Sorting
- Filtering
- Data Validation
- Conditional Formatting

LEVEL 9 — REPORTING

- Charts
- Summaries
- Dashboards
- Practical reports

LEVEL 10 — PROBLEM SOLVING

You should understand scenario instead of memorizing the formulas.

For example:

"The school wants to know how many female students in Form 2B scored above 70 in Mathematics."

Don't wait to be told to use `COUNTIFS`.

You should figure out that `COUNTIFS` is appropriate.

If you are unable to figure out what and when formula should be used, then you have learned nothing you need to repeat lessons.